

Capture the Flag 2: Widgets

Goal: Turn the single-card application into a simple multi-section interface.

Note from the instructor before you start:

These activities involve using your own problem solving and researching skills. We are strictly to implement NOT TO USE AI to generate the code for you. You may only use AI for RESEARCHING the syntax or usage, but NEVER THE CODE ITSELF.

Consequence: Immediate 0 score in your grade for this activity.

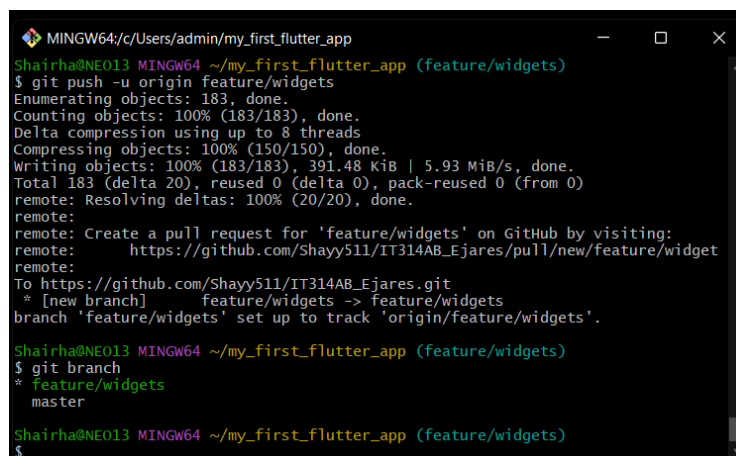
(I know what AI-generated code looks like, don't even think about it :>)

🚩 Flag 0 – Create Your Feature Branch

Objective: Create a feature branch before modifying your application. Name the branch feature/widgets

Screenshot:

- Terminal showing `feature/widgets`



```
MINGW64/c/Users/admin/my_first_flutter_app
Shairha@NE013 MINGW64 ~/my_first_flutter_app (feature/widgets)
$ git push -u origin feature/widgets
Enumerating objects: 183, done.
Counting objects: 100% (183/183), done.
Delta compression using up to 8 threads
Compressing objects: 100% (150/150), done.
Writing objects: 100% (183/183), 391.48 KiB | 5.93 MiB/s, done.
Total 183 (delta 20), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (20/20), done.
remote:
remote: Create a pull request for 'feature/widgets' on GitHub by visiting:
remote:   https://github.com/Shayy511/IT314AB_Ejares/pull/new/feature/widget
remote:
To https://github.com/Shayy511/IT314AB_Ejares.git
 * [new branch]      feature/widgets -> feature/widgets
branch 'feature/widgets' set up to track 'origin/feature/widgets'.

Shairha@NE013 MINGW64 ~/my_first_flutter_app (feature/widgets)
$ git branch
* feature/widgets
  master

Shairha@NE013 MINGW64 ~/my_first_flutter_app (feature/widgets)
$
```

🚩 Flag 1 – Break It Apart

Objective: Separate your profile information into individual `Text` widgets.

If your card currently looks like this:

```
Text(  
  
  'Chiara Marie C. Canque\n'  
  
  'BSIT - 3\n'  
  
  'My First Flutter Application\n'  
  
  'August 11, 2026',  
  
)
```

Your task: Separate each piece of information into its own `Text` widget.

You should have something conceptually like:

Text → Name

Text → Course & Section

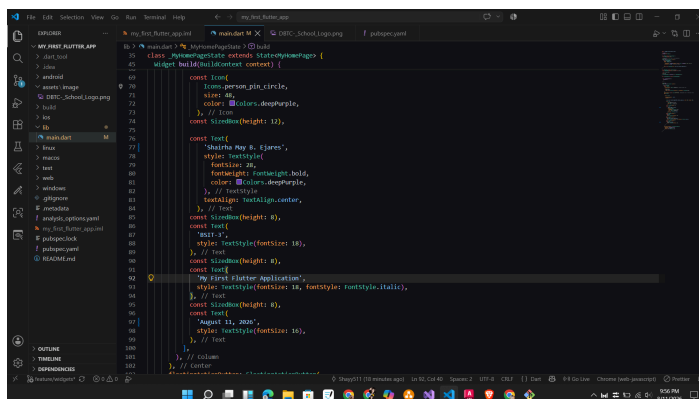
Text → Application Title

Text → Date

Do not worry about making it pretty yet.

Screenshot:

- Your code showing the separate `Text` widgets



- Your application



🚩 Flag 2 – Stack the Information

Objective: Arrange your individual `Text` widgets vertically.

Research how to use:

`Column()`

Place your information inside the `Column`.

Your card should now have:

Name

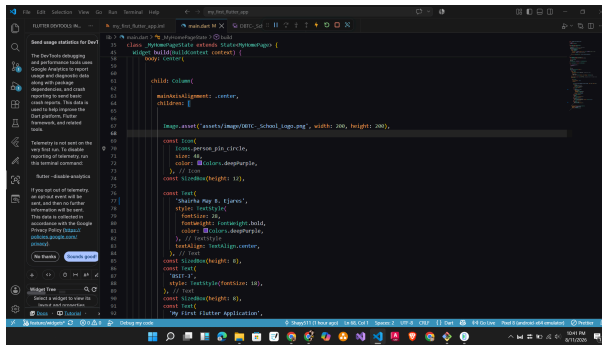
Course & Section

Application Title

Date

Screenshot:

- The **Column** in your code
- The four separate pieces of information displayed vertically



Flag 3 – Add Another Element

Objective: Add another individual widget to your **Column**.

Add something that belongs to your profile.

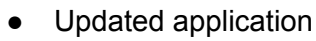
Examples:

Hobby

It must be its **own widget**, not additional text inside another **Text**.

Screenshot:

- New widget in your **Column**



Your structure should now resemble:

Column

Text

Text

Text

Text

Text

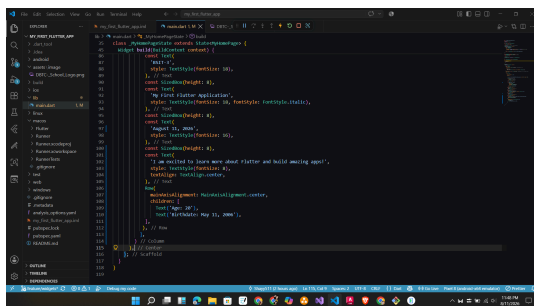
L Row

Text

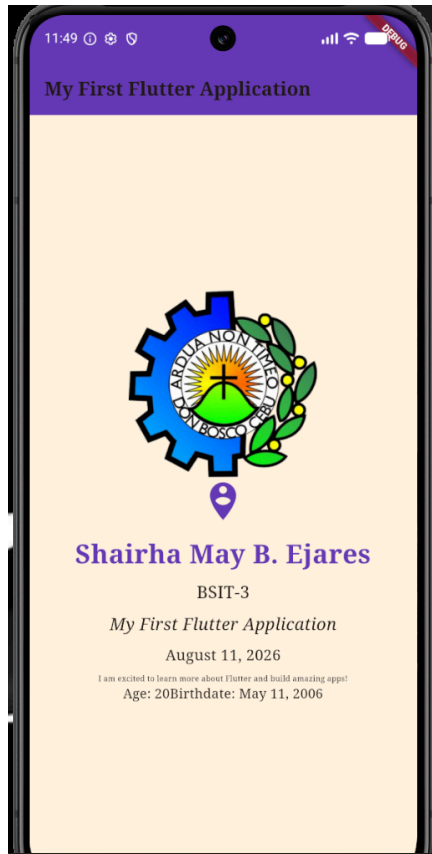
Text

Screenshot:

- Your widget structure/code



- The two items displayed side-by-side



🚩 Flag 5 – Give It Some Space

Objective: Improve the spacing between your widgets.

Research:

`EdgeInsets`

Add appropriate spacing between your:

- Image
- Name

- Course
- Other information
- Row

Challenge: Make the card easier to read **without changing its colors or fonts yet.**

Screenshot:

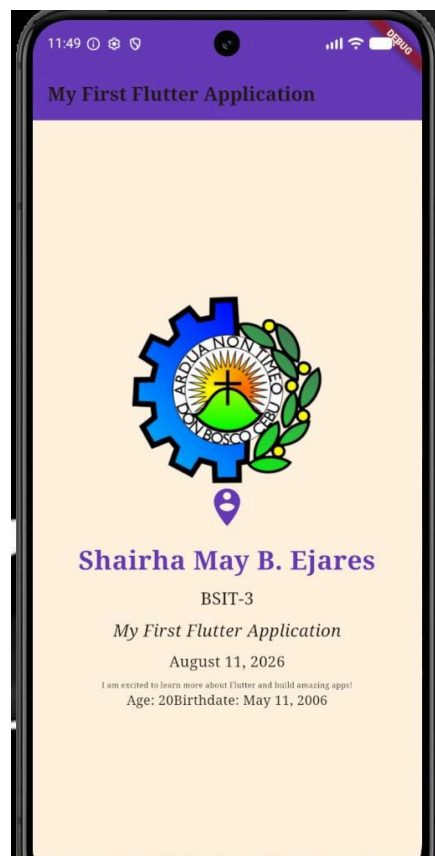
- **SizeBox** usage

```

class MyHomePageState extends State<MyHomePage> {
  widget build(BuildContext context) {
    // Text
    textAlign: TextAlign.center,
    // Text
    const SizeBox(height: 8),
    const Text(
      'TEXT-2',
      style: TextStyle(fontSize: 18),
    ), // Text
    const SizeBox(height: 8),
    const Text(
      'My First Flutter Application',
      style: TextStyle(fontSize: 18, fontStyle: FontStyle.italic),
    ), // Text
    const SizeBox(height: 8),
    const Text(
      'August 11, 2026',
      style: TextStyle(fontSize: 16),
    ), // Text
    const SizeBox(height: 8),
    const Text(
      'I am excited to learn more about Flutter and build amazing apps!',
      style: TextStyle(fontSize: 8),
      textAlign: TextAlign.center,
    ), // Text
    Row(
      mainAxisAlignment: MainAxisAlignment.center,
      children: [
        Text('Age: 20'),
        Text('Birthdate: May 11, 2006'),
      ],
    ), // Row
  ), // Column
), // Center
), // Center

```

- Updated application



🚩 Flag 6 – Make It Look Better

Objective: Apply styling to your individual widgets.

Now that your information is properly separated and positioned, improve the visual hierarchy.

Research how to modify:

`TextStyle()`

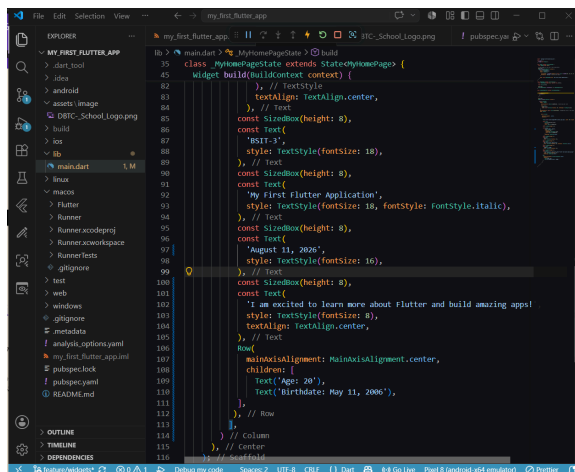
Apply different styling to at least **three** elements.

For example:

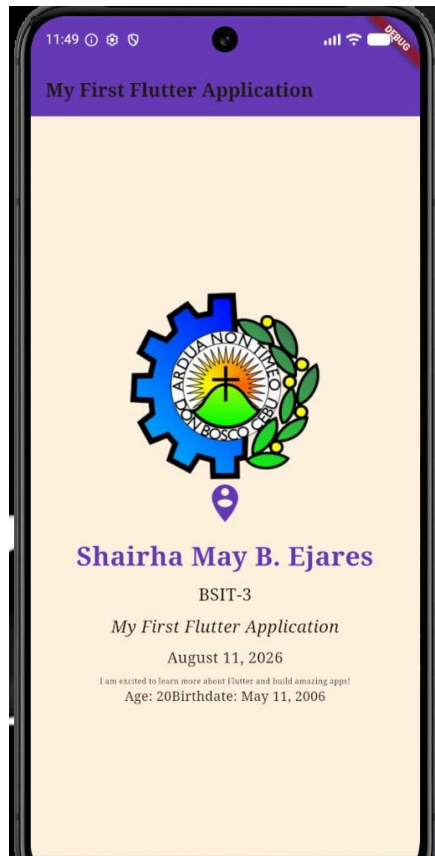
- Name → larger and bold
- Course → medium size
- Date → smaller
- Application title → italic

Screenshot:

- Your `TextStyle` changes



- Updated application



🚩 Flag 7 – Improve the Layout

Objective: Control the alignment of your widgets.

Research the properties of `Column` and `Row`.

Experiment with:

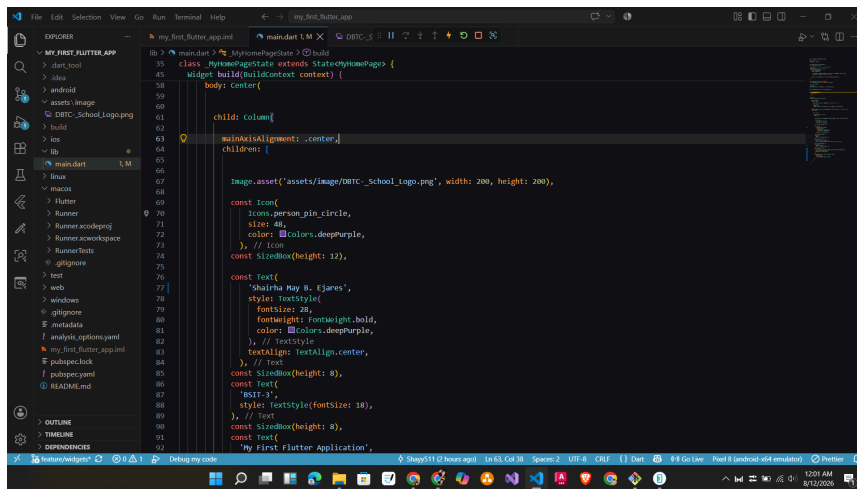
`mainAxisAlignment`

`crossAxisAlignment`

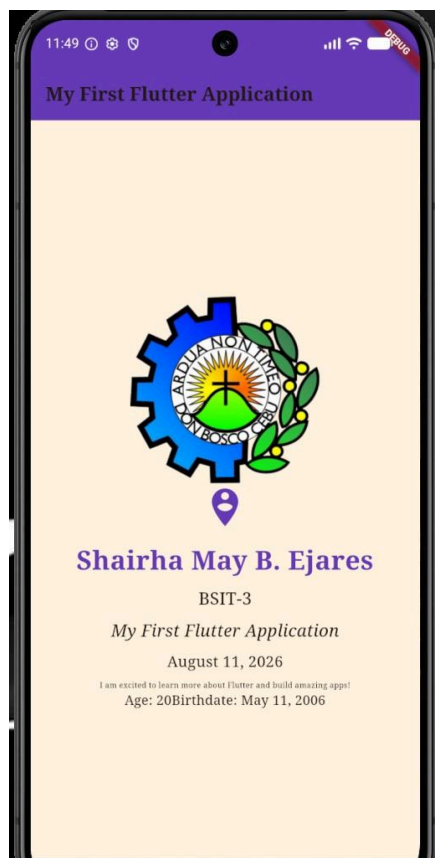
Change the alignment of your content so the card looks intentional rather than simply stacked.

Screenshot:

- The properties you changed



- Updated application



🚩 Flag 8 – Create a Second Card

Objective: Apply what you learned to create another section.

Create a second **Card** containing something different from your profile.

For example:

My Favorites

Favorite Game

Favorite Food

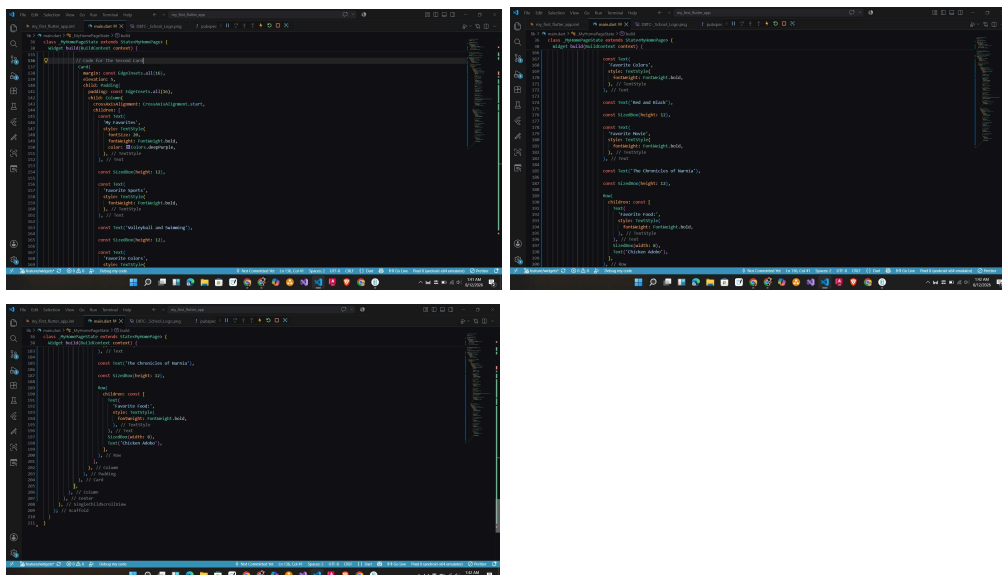
Favorite Movie

The second card should also use:

- Individual **Text** widgets
- A **Column**
- At least one **Row**
- Spacing

Screenshot:

- Code for the second card



- Both cards displayed



🚩 Flag 9 – Stack the Cards

Objective: Arrange your two cards vertically.

You now have:

Profile Card



Favorites Card

Research how to place **multiple widgets inside the Flutter body**.

Use a layout widget to arrange your two cards vertically.

Screenshot:

- The widget containing both Cards



- Both cards visible in the application

🚩 Flag 10 – Widget Discovery

Objective: Discover a Flutter widget independently.

Research the Flutter widget catalog and find **one widget you haven't used yet**.

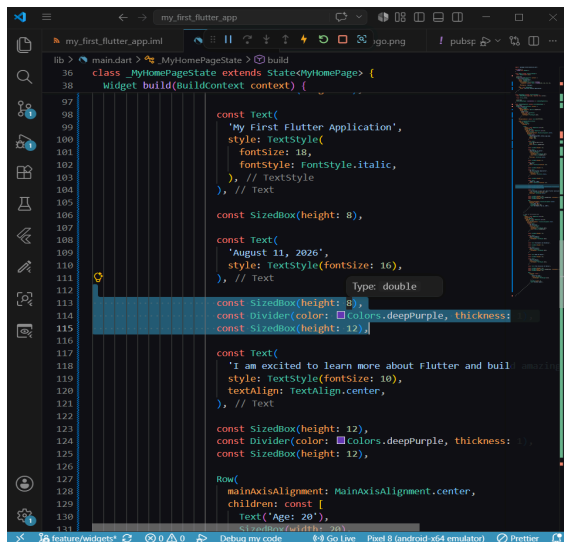
Examples:

- Icon
- Divider
- CircleAvatar
- Chip
- ElevatedButton

Add it somewhere appropriate in your application.

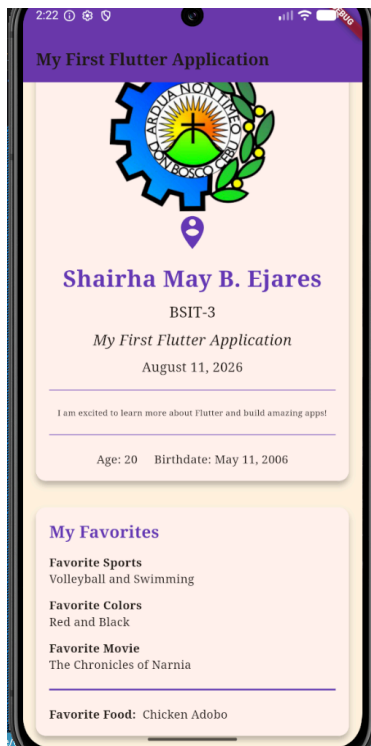
Screenshot:

- Your newly discovered widget in the code



```
lib > main.dart > MyHomePageState > build
36 class _MyHomePageState extends State<MyHomePage> {
37   Widget build(BuildContext context) {
97
98     const Text(
99       'My First Flutter Application',
100       style: TextStyle(
101         fontSize: 18,
102         fontStyle: FontStyle.italic,
103       ), // TextStyle
104     ), // Text
105
106     const SizedBox(height: 8),
107
108     const Text(
109       'August 11, 2026',
110       style: TextStyle(fontSize: 16),
111     ), // Text
112     type: double
113
114     const SizedBox(height: 8),
115     const Divider(color: Colors.deepPurple, thickness: 2),
116     const SizedBox(height: 12),
117
118     const Text(
119       'I am excited to learn more about Flutter and build amazing apps!',
120       style: TextStyle(fontSize: 10),
121       textAlign: TextAlign.center,
122     ), // Text
123
124     const SizedBox(height: 12),
125     const Divider(color: Colors.deepPurple, thickness: 2),
126     const SizedBox(height: 12),
127
128     Row(
129       mainAxisAlignment: MainAxisAlignment.center,
130       children: const [
131         Text('Age: 20'),
132         Text('Birthdate: May 11, 2006')
133       ],
134     ),
135   }
```

- The widget visible in the application



🚩 Flag 11 – Real Phone Test 📱

Objective: Run your completed application on a **physical** Android phone.

Research how to run Flutter on your phone.

Screenshot:

- Your completed application running on the physical phone (screenshot from your actual phone)



🚩 Flag 12 – Save Your Progress

Objective: Commit your completed Widget branch.

Follow proper standards of commit messages

Screenshot:

- Successful commit

```
MINGW64/c/Users/admin/my_first_flutter_app

Shairha@NE013 MINGW64 ~/my_first_flutter_app (feature/widgets)
$ git add .

Shairha@NE013 MINGW64 ~/my_first_flutter_app (feature/widgets)
$ git commit -m "Feat: Activity #2 Completed"
[feature/widgets 7120cc3] Feat: Activity #2 Completed
1 file changed, 174 insertions(+), 67 deletions(-)

Shairha@NE013 MINGW64 ~/my_first_flutter_app (feature/widgets)
$ git status
On branch feature/widgets
Your branch is ahead of 'origin/feature/widgets' by 1 commit.
(use "git push" to publish your local commits)

nothing to commit, working tree clean

Shairha@NE013 MINGW64 ~/my_first_flutter_app (feature/widgets)
$ git push origin feature/widgets
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (4/4), 1.53 KiB | 522.00 KiB/s, done.
```

- feature/widgets branch

